

Customer Churn Analysis

Final Data Science Project Submission

 github.com/ahsannoor12/Final-Project

 Kaggle Telco Churn Dataset

PROJECT SUMMARY

"This project predicts customer churn using the Telco dataset. By applying Random Forest, I identified key retention factors to improve business outcomes."

Data Collection & Source

We utilized the Telco Customer Churn dataset from Kaggle to solve the real-world problem of subscriber attrition.

-  **7,043** unique customer records.
-  **21** demographic and service variables.
-  Cleaned 'TotalCharges' for accurate analysis.

Dataset Link: [kaggle.com/telco-churn](https://www.kaggle.com/datasets/wcaesare/telco-churn)



Project Workflow

Cleaning

Imputed null values and removed irrelevant IDs to ensure data integrity.

EDA

Visualized relationships between contract type, tenure, and churn rates.

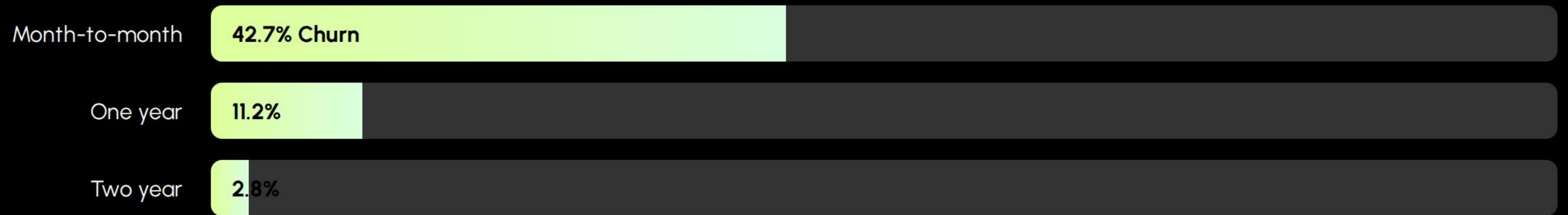
Engineering

Scaled numerical data and encoded categorical variables for the model.

Modeling

Applied Random Forest algorithm to predict customer departure likelihood.

Churn by Contract Type



Observation: Month-to-month contracts are high-risk segments needing retention focus.

Model Accuracy

80%

Test Results

The optimized Random Forest model achieved 80% accuracy in identifying potential churners.

This allows the marketing team to proactively reach out to at-risk subscribers before they leave.

Business Recommendations

-  Incentivize Month-to-Month customers to switch to long-term contracts.
 -  Improve technical support for Fiber Optic users to reduce dissatisfaction.
 -  Focus retention efforts on customers with less than 6 months of tenure.
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Final Deliverables

 Repository: [ahsannoor12/Final-Project](#)

Dataset: Telco Customer Churn (Kaggle)

Thank You!

Image Sources

Thumbnail https://assets.qlik.com/image/upload/w_1720/q_auto/qlik/glossary/dashboard-examples/seo-financial-dashboards-cfo-dashboard_f9frfk.png
for Source: www.qlik.com
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